Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/04/2020
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Early Stage Plant Adaptation to Spaceflight - Molecular Responses of Arabidopsis to the Transition from Terrestrial Environment to Space		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/31/2020 per NSSC information (Ed., 11/13/19) NOTE: End date changed to 10/31/2019 per NSSC information (Ed., 10/12/18) NOTE: End date changed to 10/31/2018 per NSSC information (Ed., 12/13/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Paul, Anna-Lisa Ph.D. (University of Florida)		
Grant/Contract No.:	NNX15AB12G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this proposal is to use suborbital spaceflights in SpaceShipTwo to develop an understanding of the genomic and metabolic plant responses upon entry into spaceflight. The long term goal of this line of research is a comprehensive appreciation of the biological responses that occur as terrestrial life experiences phases and aspects of spaceflight. We will build on our previous genomic work using Arabidopsis plants by imaging green fluorescent protein (GFP)-tagged promoters from specific genes indicated as potentially involved in early spaceflight transitions and by fixing plants during suborbital flight for whole genome expression analysis with arrays and RNASeq. This project makes extensive use of a unique middeck locker imager system that was developed within the NASA Flight Opportunities Program, an imager that is capable of delivering multispectral fluorescent images at high resolution within space vehicles. The project also utilizes the specialized experience of the Principal Investigator (PI) team in deploying plant molecular biology gene expression experiments and imaging in diverse space-related environments. These studies are innovative in that they will provide the first molecular genetic data from suborbital spaceflights, thereby filling a substantial gap in knowledge regarding the early stages and temporal dynamics of gene expression responses to spaceflight. Specifically these data will provide insights into the effects of early stage spaceflight responses so as to begin to round out a more comprehensive understanding of the temporal and transitional plant responses to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research Impact/Earth Benefits: To gain fundamental insight into how plants perceive and process the signals that control tropism and adaptive physiology in novel and extreme environments. It also provides a unique understanding of how a terrestrial organism transitions from a normal gravity environment to one outside its evolutionary experience. Currently only suborbital vehicles can facilitate access to this unique window of transition for data collection. This research also highlights how suborbital vehicles can be used to optimize experimental design to help maximize International Space Station (ISS) utilization in future experiments.
Task Progress:	2018-2019 Flight Campaigns During the past project year the FLEX imager has flown two sub-orbital missions with successful results. In December 2018, the FLEX unit “Bullwinkle” was deployed to the Virgin Galactic Spaceship 2 for a suborbital mission. This flight was the first experimental flight with hardware flown by Virgin Galactic, and the result was successful collection of biological data. In January 2019, the FLEX unit “Rocky” was installed on the Blue Origin New Shepard rocket capsule and returned flight images and flight profile data. Flight Biology: Two types of plates were flown in the FLEX imager during suborbital flight. The FLEX GFP images and thermal (FLIR) plate consisted of 17-20 day old plants. These plates were horizontally grown to allow the most leaf area possible for study. Two different seed lines were flown, a wild-type Col-0 line and a knock out mutant (background Col-0) line that has been shown to been drought tolerant. The GFP plate that was flow in suborbital studies consisted of 5 different environmental signalling genotypes that were flown at 6-7 days old. Three of them have been grown and studied aboard the International Space Station (ISS), and two are Ca2+ regulator lines that may help determine plants minute and instantaneous response to the microgravity environment. Flight Environments Virgin Galactic: The Virgin Galactic (VG) flight environment was climate controlled late in the pre-launch timeline, but there was limited PI access after the payload loaded for launch. Sample installation was expertly done by the VG engineer, but would have been much better to have the final calibration checks conducted by the PIs just prior to launch. Retrieval was as rapid as possible, and the VG support group did a great job of enabling the very best possible scenario, but faster would be better in future experiments. Blue Origin: The Blue Origin original climate control was outside of experimental parameters (10-35 Degrees Celsius), with internal measurements of the testing materials temperature as low as 4.5 deg C. The samples were reflown. During the re-flight on Blue Origin, biochemistry plates were flown. These plates were able to be harvested within about 30 minutes of landing by the PI during the field recovery period. Flight Engineering Outcomes: The testing of both imagers was successful on their maiden flights. The NASA Flight Opportunities program objective of the flights was the technology maturation of the FLEX imaging system. The imagers, developed in parallel, to serve as flight and ground control units were both utilized as flight-units due to launch dates and mission parameters. This setup allowed the non-flight imager to still be used as the ground control unit, while the flight imager could be individualized to the needs of the flight provider. The cold conditions of the Blue Origin flight made aspects of the flight difficult to qualify. The temperatures were outside of the planned ranged for the biology and therefore we are unable to determine the biological significance, especially of thermal imaging during the flight. However, the payload was provided a reflight, which kept the temperature about 10 degrees Celsius during the flight. During the re-flight on Blue Origin the monochrome GFP camera did not work as expected. This seems to be a reoccurring theme with Blue Origin that while all programs and systems appear fine, once they interact with the flight capsule they seem to encounter glitches. We are still working to understand the phenomenon to be able to better protect against it in the future. Hardware Results: High quality FLEX GFP images and thermal (FLIR) images were recovered during maiden flights on both Blue Origin and Virgin Galactic flights and thermal data on the Blue Origin re-flight. Flight data tracked using the MSR program was collected for the Blue Origin flight. Blue Origin Re-flight: Due to mission parameters during flight falling outside of expected temperature, Flight Opportunities and Blue Origin agreed to re-fly the payload. The flight flew December 2019, on Blue Origin flight P9. Excellent thermal imaging photographs were collected, and gene expression data were collected from the post-flight samples. Blue Origin and VG Biology Biochemical Analysis: Due to the speedy recovery of the biological samples during the Blue Origin December 2019 flight, and the previous Virgin Galactic 2018 flight, material was available for transcriptome sequencing. Ground control samples were harvested in parallel for ground/flight comparisons. Material was harvested onsite and stored in RNALater. Upon return to the UF Space Plants lab, samples were stored at -80 degrees Celsius.

	For each flight, the same wild-type ecotype was selected. For the 2018 Virgin Galactic flight, the WS ecotype was selected. For the Dec. 2019 flight, WS and a Sku5 knock-out mutant was selected. These selections were based on the primary investigators' experience with the seed lines in orbital and ground-based studies. RNA was extracted from the flight and ground samples using the Qiagen RNAeasy Plant kit. These samples were then sequenced using the UF Interdisciplinary Center for Biotechnology Research. Results were received at the end of February 2020 and analysis is ongoing. Conclusions This successfully completes the Early Adaptation to Spaceflight project. In summary, the project completed three successful sub-orbital flights, returning data from each for analysis. Ongoing study of the RNA-Seq Biochemical analysis, thermal response video, and calcium/protein GFP movement all show early response to the sub-orbital flight environment, though final analysis is still ongoing. The responses to novel environments are crucial to not only future off Earth plant health but to understanding and decoding pathways in plants here on Earth. Both scientific and engineering avenues of study in this report are still open and ongoing. Data is still being generated on leaf temperature data from the FLIR thermal videos. The thermal modeling is currently being worked to consider the data we have from the suborbital flights to help with future accuracy. The monochrome camera files that were used to record GFP protein responses are currently being used to help select plants seed lines for future flights, and to help better understand high resolution photography during its transition to space. Finally, the FLEX imager is currently being upgraded with the latest technology and using the lessons learned from past flights to help build more robust technology for use on future sub-orbital flights.
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Califar B, Zupanska A, Callaham JA, Bamsey MT, Graham T, Paul A-L, Ferl RJ. "Shared metabolic remodeling processes characterize the transcriptome of Arabidopsis thaliana within various suborbital flight environments." Gravit Space Res. 2021 Jan 28;9(1):13-29. https://doi.org/10.2478/gsr-2021-0002 , Jan-2021